

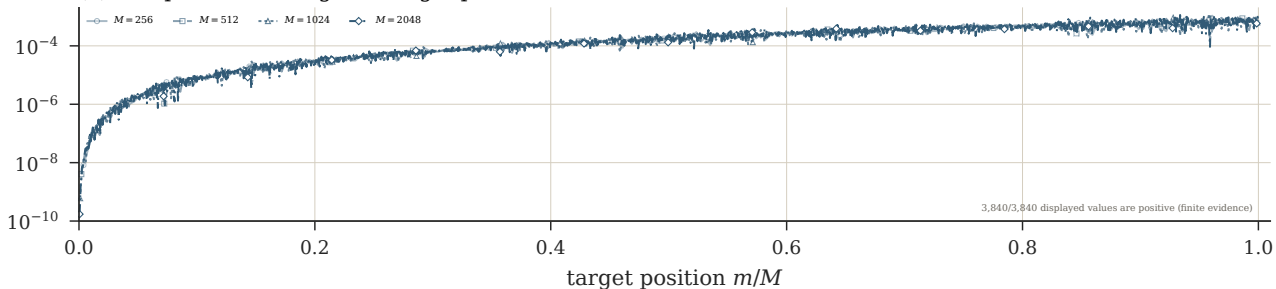


Heat-weighted quartic profiles and the unweighted correlation gap

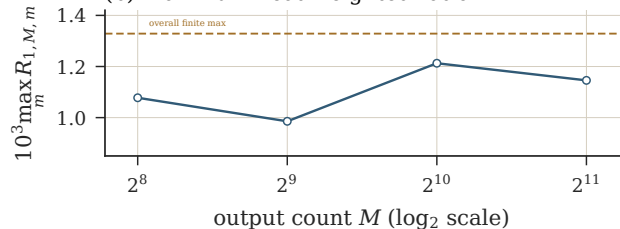
All-index theorem: $|R_{L,M,m}| \leq 7.8343 (m/M)^2 \sqrt{M}$ · finite extension: 4,042 triples and 27.082 billion ordered paths

normalized ratio $R_{L,M,m}$ (log scale)

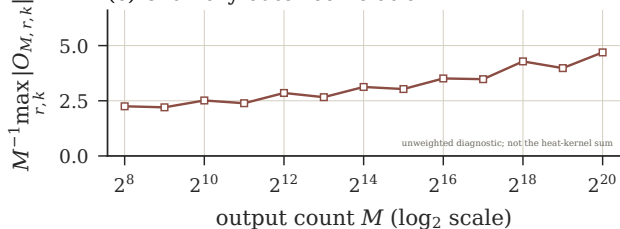
(a) Complete heat-weighted target profiles, $L = 1$



(b) Maximum heat-weighted ratio



(c) Ordinary outer correlation



Interpretation: the proved ceiling removes all L -growth but leaves $\sqrt{M}$. Panel (c) shows why a supremum bound on the heat kernel is too crude; the next lemma must retain smooth heat weights and time ordering.